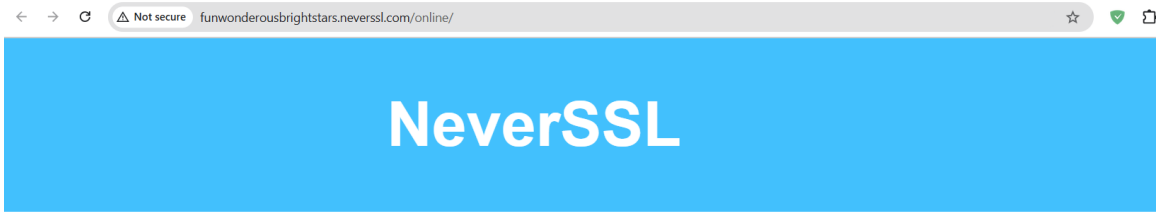
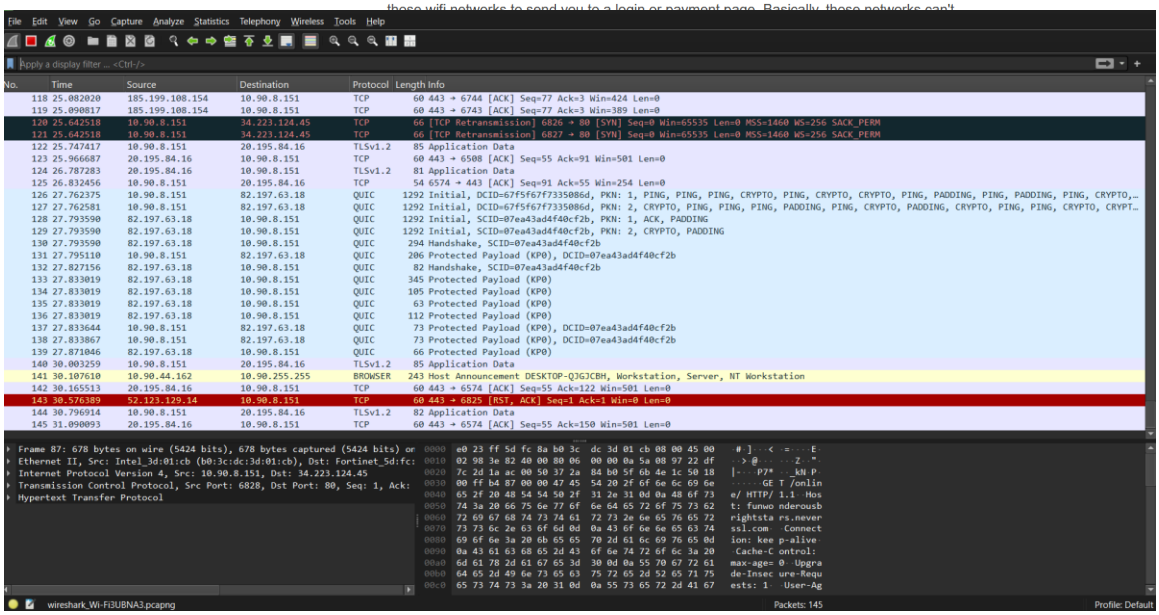
Part1

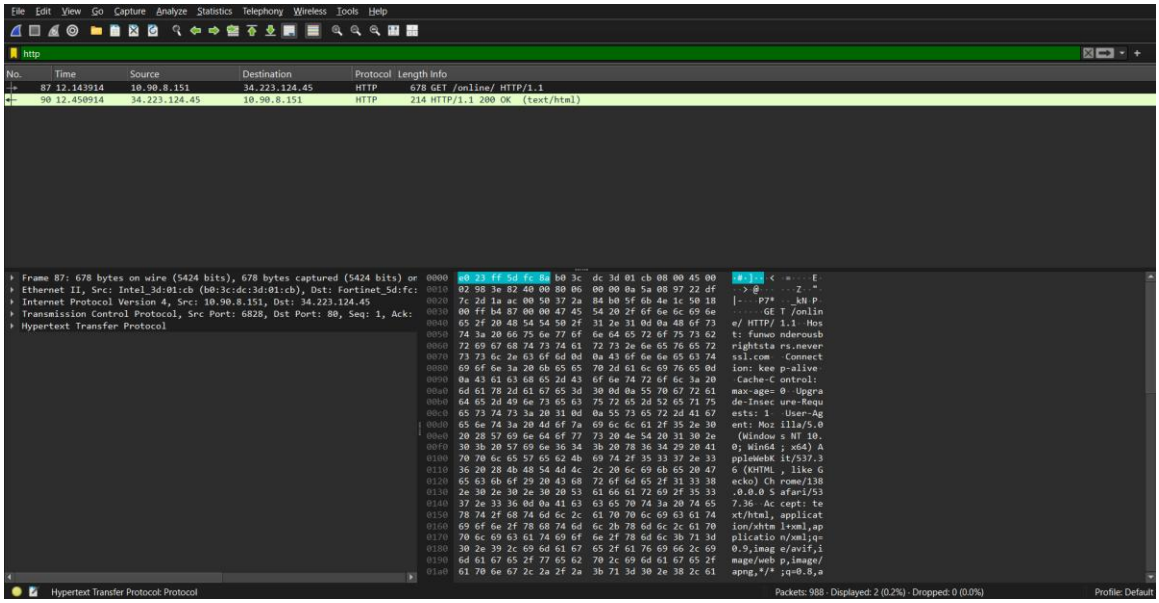
task1


Not secure
funwonderousbrightstars.neverssl.com/online/
NeverSSL
What?
This website is for when you try to open Facebook, Google, Amazon, etc on a wifi network, and nothing happens. Type "http://neverssl.com" into your browser's url bar, and you'll be able to log on.
How?
neverssl.com will never use SSL (also known as TLS). No encryption, no strong authentication, no HSTS, no HTTP/2.0, just plain old unencrypted HTTP and forever stuck in the dark ages of internet security.
Why?
Normally, that's a bad idea. You should always use SSL and secure encryption when possible. In fact, it's such a bad idea that most websites are now using https by default.
And that's great, but it also means that if you're relying on poorly-behaved wifi networks, it can be hard to get online. Secure browsers and websites using https make it impossible for


File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
118 25.082020 185.199.108.154 10.90.8.151 TCP 60 443 → 6744 [ACK] Seq=77 Ack=3 Win=424 Len=0
119 25.090817 185.199.108.154 10.90.8.151 TCP 60 443 → 6743 [ACK] Seq=77 Ack=3 Win=389 Len=0
120 25.642518 10.90.8.151 34.223.124.45 TCP 66 [TCP Retransmission] 6826 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
121 25.642518 10.90.8.151 34.223.124.45 TCP 66 [TCP Retransmission] 6827 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
122 25.747417 10.90.8.151 20.195.84.16 TLSv1.2 85 Application Data
123 25.966687 20.195.84.16 10.90.8.151 TCP 60 443 → 6508 [ACK] Seq=55 Ack=91 Win=501 Len=0
124 26.787283 20.195.84.16 10.90.8.151 TLSv1.2 81 Application Data
125 26.832456 10.90.8.151 20.195.84.16 TCP 54 6574 → 443 [ACK] Seq=91 Ack=55 Win=254 Len=0
126 27.762375 10.90.8.151 82.197.63.18 QUIC 1292 Initial, DCID=67f5f67f733508d, PKN: 1, PING, PING, PING, CRYPTO, PING, CRYPTO, CRYPTO, PING, PADDING, PING, PADDING, PING, CRYPTO,…
127 27.762581 10.90.8.151 82.197.63.18 QUIC 1292 Initial, DCID=67f5f67f733508d, PKN: 2, CRYPTO, PING, PING, PING, PADDING, PING, CRYPTO, PADDING, CRYPTO, PING, PING, CRYPTO, CRYPT…
128 27.793590 82.197.63.18 10.90.8.151 QUIC 1292 Initial, SCID=07ea43ad4f40cf2b, PKN: 1, ACK, PADDING
129 27.793590 82.197.63.18 10.90.8.151 QUIC 1292 Initial, SCID=07ea43ad4f40cf2b, PKN: 2, CRYPTO, PADDING
130 27.793590 82.197.63.18 10.90.8.151 QUIC 294 Handshake, SCID=07ea43ad4f40cf2b
131 27.795110 10.90.8.151 82.197.63.18 QUIC 206 Protected Payload (KP0), DCID=07ea43ad4f40cf2b
132 27.827156 82.197.63.18 10.90.8.151 QUIC 82 Handshake, SCID=07ea43ad4f40cf2b
133 27.833019 82.197.63.18 10.90.8.151 QUIC 345 Protected Payload (KP0)
134 27.833019 82.197.63.18 10.90.8.151 QUIC 105 Protected Payload (KP0)
135 27.833019 82.197.63.18 10.90.8.151 QUIC 63 Protected Payload (KP0)
136 27.833019 82.197.63.18 10.90.8.151 QUIC 112 Protected Payload (KP0)
137 27.833644 10.90.8.151 82.197.63.18 QUIC 73 Protected Payload (KP0), DCID=07ea43ad4f40cf2b
138 27.833867 10.90.8.151 82.197.63.18 QUIC 73 Protected Payload (KP0), DCID=07ea43ad4f40cf2b
139 27.871046 82.197.63.18 10.90.8.151 QUIC 66 Protected Payload (KP0)
140 30.003259 10.90.8.151 20.195.84.16 TLSv1.2 85 Application Data
141 30.107610 10.90.44.162 10.90.255.255 BROWSER 243 Host Announcement DESKTOP-Q3GJCBH, Workstation, Server, NT Workstation
142 30.165513 20.195.84.16 10.90.8.151 TCP 60 443 → 6574 [ACK] Seq=55 Ack=122 Win=501 Len=0
143 30.576389 52.123.129.14 10.90.8.151 TCP 60 443 → 6825 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
144 30.796914 10.90.8.151 20.195.84.16 TLSv1.2 82 Application Data
145 31.090093 20.195.84.16 10.90.8.151 TCP 60 443 → 6574 [ACK] Seq=55 Ack=150 Win=501 Len=0
Frame 87: 678 bytes on wire (5424 bits), 678 bytes captured (5424 bits) or
Ethernet II, Src: Intel_3d:01:cb (b0:3c:dc:3d:01:cb), Dst: Fortinet_5d:fc:
Internet Protocol Version 4, Src: 10.90.8.151, Dst: 34.223.124.45
Transmission Control Protocol, Src Port: 6828, Dst Port: 80, Seq: 1, Ack:
Hypertext Transfer Protocol
wireshark_Wi-Fi3UBNA3.pcapng
Packets: 145
Profile: Default

TASK 2

pp



Part 2

TCP/IP

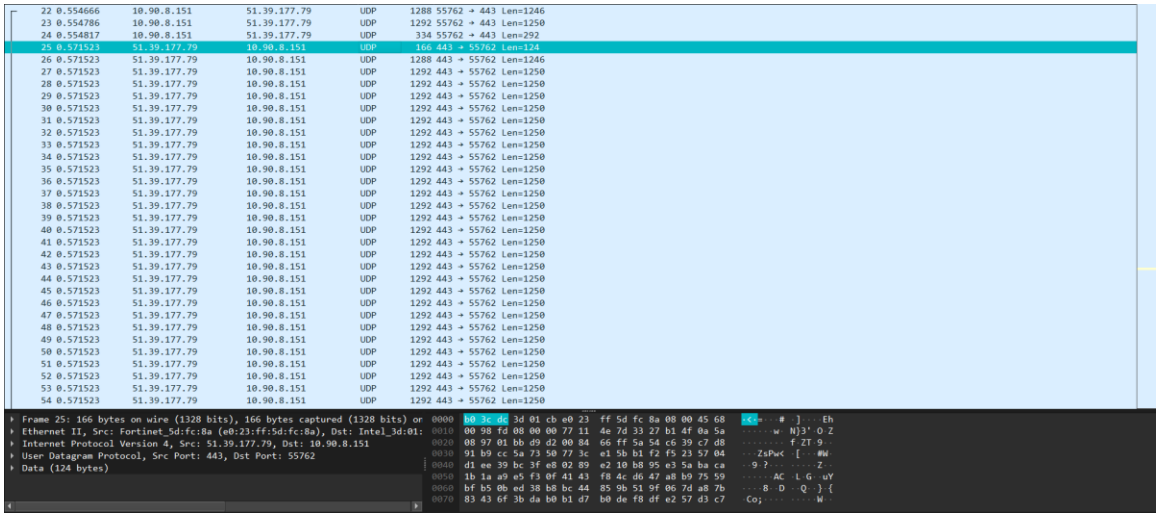

Info Length Protocol Destination Source Time
Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM [SYN] 80 → 54102 86 TCP …:2600:1f13:37c:1400 …2001:16a2:c01d:e5b4 57.215692 723
54102 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM [TCP Retransmission] 86 TCP …:2600:1f13:37c:1400 …2001:16a2:c01d:e5b4 58.215579 773
54102 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM [TCP Retransmission] 86 TCP …:2600:1f13:37c:1400 …2001:16a2:c01d:e5b4 60.221946 814
54102 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM [TCP Retransmission] 86 TCP …:2600:1f13:37c:1400 …2001:16a2:c01d:e5b4 64.232894 855
Seq=0 Ack=1 Win=26140 Len=0 MSS=1307 [SYN, ACK] 54102 → 80 78 TCP …2001:16a2:c01d:e5b4 …:2600:1f13:37c:1400 64.520436 865
Seq=1 Ack=1 Win=65535 Len=0 [ACK] 80 → 54102 74 TCP …:2600:1f13:37c:1400 …2001:16a2:c01d:e5b4 64.520805 866
GET /online HTTP/1.1 579 HTTP …:2600:1f13:37c:1400 …2001:16a2:c01d:e5b4 64.522159 867
Seq=1 Ack=506 Win=26800 Len=0 [ACK] 54102 → 80 74 TCP …2001:16a2:c01d:e5b4 …:2600:1f13:37c:1400 64.861554 871
HTTP/1.1 301 Moved Permanently (text/html) 631 HTTP …2001:16a2:c01d:e5b4 …:2600:1f13:37c:1400 64.871188 872
GET /online/ HTTP/1.1 580 HTTP …:2600:1f13:37c:1400 …2001:16a2:c01d:e5b4 64.887358 876
Seq=558 Ack=1012 Win=27872 Len=1300 [TCP PDU reassembled in 890] [ACK] 54102 → 80 1374 TCP …2001:16a2:c01d:e5b4 …:2600:1f13:37c:1400 65.252196 889
HTTP/1.1 200 OK (text/html) 293 HTTP …2001:16a2:c01d:e5b4 …:2600:1f13:37c:1400 65.252196 890
Seq=1012 Ack=2077 Win=65535 Len=0 [ACK] 80 → 54102 74 TCP …:2600:1f13:37c:1400 …2001:16a2:c01d:e5b4 65.252366 893
GET /favicon.ico HTTP/1.1 522 HTTP …:2600:1f13:37c:1400 …2001:16a2:c01d:e5b4 65.360536 896
HTTP/1.1 200 OK (PNG) 490 HTTP …2001:16a2:c01d:e5b4 …:2600:1f13:37c:1400 65.736007 898
Seq=1460 Ack=2493 Win=65119 Len=0 [ACK] 80 → 54102 74 TCP …:2600:1f13:37c:1400 …2001:16a2:c01d:e5b4 65.787617 900
0000 bc 03 58 ee 09 df a2 fc a7 56 e4 bc 86 dd 60 00 ··X····· ·V····`·
0010 00 00 02 41 06 f2 26 00 1f 13 03 7c 14 00 ba 21 ···A··&· ···|···!
0020 71 65 5f c7 73 6e 20 01 16 a2 c0 1d e5 b4 54 58 qe_·sn ·· ·····TX
0030 e7 7c d1 d6 e4 82 00 50 d3 56 0e 75 41 0f 9e 98 ·|·····P ·V·uA···
0040 05 8d 50 18 68 b0 12 1a 00 00 48 54 54 50 2f 31 ··P·h··· ··HTTP/1
0050 2e 31 20 33 30 31 20 4d 6f 76 65 64 20 50 65 72 .1 301 M oved Per
0060 6d 61 6e 65 6e 74 6c 79 0d 0a 44 61 74 65 3a 20 manently ··Date:
0070 54 68 75 2c 20 32 36 20 4a 75 6e 20 32 30 32 35 Thu, 26 Jun 2025
0080 20 31 36 3a 34 30 3a 33 30 20 47 4d 54 0d 0a 53 16:40:3 0 GMT··S
0090 65 72 76 65 72 3a 20 41 70 61 63 68 65 2f 32 2e erver: A pache/2.
00a0 34 2e 36 32 20 28 29 0d 0a 4c 6f 63 61 74 69 6f 4.62 () ··Locatio
00b0 6e 3a 20 68 74 74 70 3a 2f 2f 73 75 62 6c 69 6d n: http: //sublim
00c0 65 73 70 6c 65 6e 64 69 64 74 72 61 6e 73 63 65 esplendi dtransce
00d0 6e 64 65 6e 74 6b 69 73 73 2e 6e 65 76 65 72 73 ndentkis s.nevers
00e0 73 6c 2e 63 6f 6d 2f 6f 6e 6c 69 6e 65 2f 0d 0a sl.com/o nline/··
00f0 43 6f 6e 74 65 6e 74 2d 4c 65 6e 67 74 68 3a 20 Content- Length:
0100 32 36 37 0d 0a 4b 65 65 70 2d 41 6c 69 76 65 3a 267··Kee p-Alive:
0110 20 74 69 6d 65 6f 75 74 3d 35 2c 20 6d 61 78 3d timeout =5, max=
0120 31 30 30 0d 0a 43 6f 6e 6e 65 63 74 69 6f 6e 3a 100··Con nection:
0130 20 4b 65 65 70 2d 41 6c 69 76 65 0d 0a 43 6f 6e Keep-Al ive··Con
0140 74 65 6e 74 2d 54 79 70 65 3a 20 74 65 78 74 2f tent-Typ e: text/
0150 68 74 6d 6c 3b 20 63 68 61 72 73 65 74 3d 69 73 html; ch arset=is
ured (5048 bits) on interface \Device\NPF_{DEB47314-1837-4FEC-8138-0FAB50CE8678}, id 0
I, Src: a2:fc:a7:56:e4:bc (a2:fc:a7:56:e4:bc), Dst: Intel_ee:09:df (bc:03:58:ee:09:df)
: 2600:1f13:37c:1400:ba21:7165:5fc7:736e, Dst: 2001:16a2:c01d:e5b4:5458:e77c:d1d6:e482
ansmission Control Protocol, Src Port: 80, Dst Port: 54102, Seq: 1, Ack: 506, Len: 557
Hypertext Transfer Protocol
Line-based text data: text/html (7 lines)
GET /online HTTP/1.1
Host: sublimesplendidtranscendentkiss.neverssl.com
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/137.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Referer: http://neverssl.com/
Accept-Encoding: gzip, deflate
Accept-Language: ar,en-US;q=0.9,en;q=0.8
HTTP/1.1 301 Moved Permanently
Date: Thu, 26 Jun 2025 16:40:30 GMT
Server: Apache/2.4.62 ()
Location: http://sublimesplendidtranscendentkiss.neverssl.com/online/
Content-Length: 267
Keep-Alive: timeout=5, max=100
Connection: Keep-Alive
Content-Type: text/html; charset=iso-8859-1
<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>301 Moved Permanently</title>
</head><body>
<h1>Moved Permanently</h1>
The document has moved here.
</body></html>
GET /online/ HTTP/1.1
Host: sublimesplendidtranscendentkiss.neverssl.com
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/137.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Referer: http://neverssl.com/
Accept-Encoding: gzip, deflate
Stream No delta times ASCII Show as Entire conversation (3951 bytes)
Find Next Case sensitive Find:

```
GET /online/ HTTP/1.1
Host: sublimesplendidtranscendentkiss.neverssl.com
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/137.0.0.0 Saf
ari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,applicat
ion/signed-exchange;v=b3;q=0.7
Referer: http://neverssl.com/
Accept-Encoding: gzip, deflate
Accept-Language: ar,en-US;q=0.9,en;q=0.8

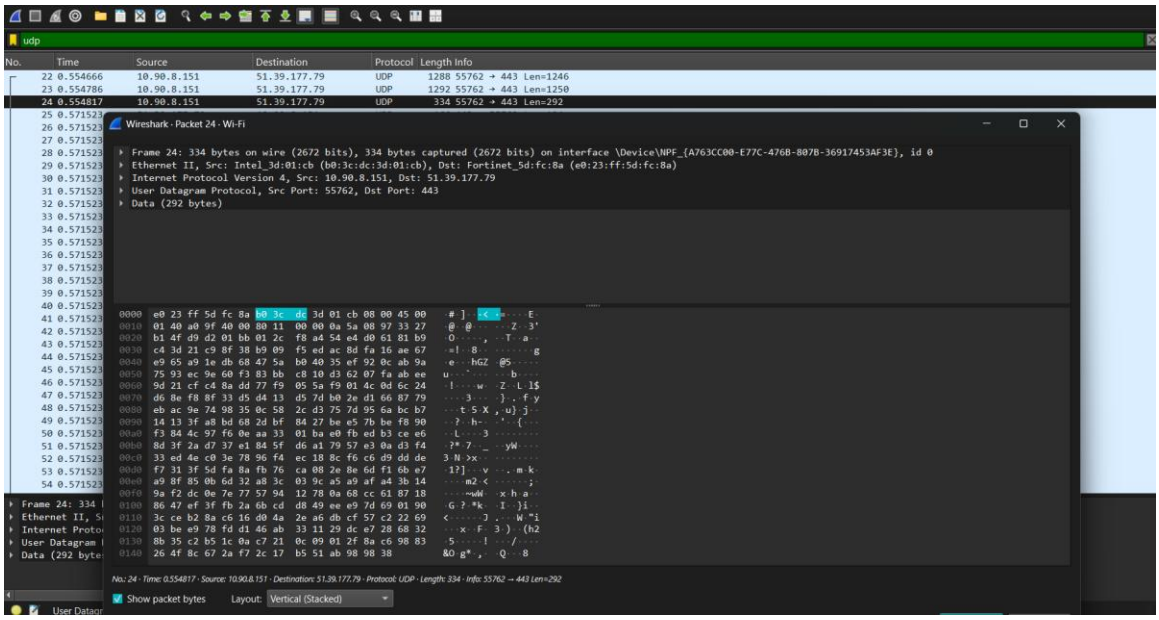

HTTP/1.1 200 OK
Date: Thu, 26 Jun 2025 16:40:31 GMT
Server: Apache/2.4.62 ()
Last-Modified: Wed, 29 Jun 2022 00:23:22 GMT
ETag: "8be-5e28b29291e10-gzip"
Accept-Ranges: bytes
Vary: Accept-Encoding
Content-Encoding: gzip
Content-Length: 1173
Keep-Alive: timeout=5, max=99
Connection: Keep-Alive
Content-Type: text/html; charset=UTF-8
```

part3

task 1



task 2



TCP or UDP	Reasons
Reliability and Connection Establishment	**TCP**: Reliable, uses a three-way handshake (SYN, SYN-ACK, ACK) to establish a connection. Retransmits lost packets. **UDP**: Unreliable, no prior connection or retransmission of lost packets.
Data Integrity and Ordering	**TCP**: Ensures data integrity with checksums and orders packets using sequence numbers. **UDP**: Includes a basic checksum but does not guarantee ordering or retransmission of lost packets.

TCP	UDP
Use Cases	Web browsing (HTTP/HTTPS), email (SMTP, IMAP), file transfers (FTP). Suitable for applications requiring reliable, ordered data delivery.
Performance	Slower due to handshake, retransmissions, and congestion control. Higher overhead from larger headers and reliability mechanisms.